Pre-scribe

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Summary

- **Pre-Scribe** is a machine learning solution designed to solve the $16.5B student attrition crisis in higher education. By shifting from a reactive "end-of-semester" review process to a proactive "early-warning" model, **Pre-Scribe** identifies at-risk students within the first four weeks of the academic year. Utilizing 37 key demographic and socio-economic indicators from the UCI Student Dropout dataset, the system generates a "Dropout Risk Probability Score" for every student. This allows academic advisors to "prescribe" targeted financial and academic interventions before a student withdraws, effectively protecting institutional revenue and ensuring student persistence.

Data source

- **Primary Source:** UCI Machine Learning Repository: Predict Students' Dropout and Academic Success.
  - **Metadata:** dataset consists of 4,424 records, with 37 attributes captured at enrollment and through the first two semesters:
  - ✓ **Demographics:** Nationality, gender, marital status and age at enrollment.
  - ✓ **Socio-economics:** Scholarship holder status, debtor status, and parental qualifications.
  - ✓ **Academic Performance:** Curricular units enrolled, approved, and grades for semesters 1 and 2.
  - ✓ **Target:** `Dropout`, `Enrolled`, or `Graduate`.

- **Stakeholders:**
  - **University Provosts/CFOs:** Focused on the tuition recovery and graduation rate.
  - **Academic Advisors:** The primary users of the "Early Warning" alerts to trigger interventions.
  - **Students:** The ultimate beneficiaries of timely support and resource allocation.
  - **Data Science Mentor:** Responsible for auditing the technical integrity of the DSM approach.

Business goal

SMART GOAL: To develop and validate a machine learning model that predicts student dropout risk with at least **80% Recall** by the end of the first semester, enabling academic advisors to conduct targeted interventions for at-risk students and potentially recovering **$500k–$1M** in annual tuition revenue for a mid-sized institution.

- **Specific:** Target student dropout using 37 demographic, socio-economic, and academic features.
- **Measurable:** Success is defined by a Recall score of >0.80 and F1-score of >0.75.
- **Achievable:** Using the expert-curated UCI dataset, with no missing values.
- **Relevant:** Directly addresses the primary financial pain point (tuition loss) of university stakeholders.
- **Time-bound:** Completion within the 8-week Capstone 2 timeframe.

Criteria for Success

- **Technical Metric:** Achieve a high **Recall** (Sensitivity). In this business case, a "False Negative" (missing a student who drops out) is more costly than a "False Positive" (offering extra help to a student who might have stayed).
- **Business Metric:** Identify the top 3 **leading indicators** of dropout (e.g., "Debtor status") to inform institutional policy.
- **Ethics Statement:** This project will prioritize student privacy and **supportive intervention**. The intent is to allocate more resources to identified at-risk students, not to serve as punitive measures.
- **Deliverable example:** A lead-scoring dashboard that ranks students by "Dropout Risk Probability."

Scope & Constraints

- **In-Scope:** Analysis of 37 features including enrollment data, socio-economic backgrounds, and academic performance from the first two semesters. Training and testing of classification algorithms (Random Forest, XGBoost).
- **Out-of-Scope:** Real-time integration with a live University Registrar system; analysis of qualitative data (e.g., student interview transcripts or social media sentiment).
- **Ethical Constraints:** The model must be audited to ensure it does not unfairly penalize students based on sensitive demographic attributes (gender, nationality).
- **Data Constraints:** The project is limited to the features provided in the UCI dataset; it cannot account for external "black swan" events (e.g., a sudden family emergency not reflected in the data).

Data wrangling

- **Standardization:** Categorical variables (e.g., Marital Status, Course) will be converted into consistent numeric or dummy-coded formats.
- **Target Transformation:** The `Target` variable will be re-encoded into a binary format: **1 for Dropout** and **0 for Retained** (merging 'Graduate' and 'Enrolled').
- **Integrity Check:** A standard `df.isnull().sum()` check will be performed to confirm the absence of missing values.
- **Feature Filtering:** To maintain the "Early Warning" promise, I will remove features that are only known *after* a student leaves (e.g., 2nd-semester grades), ensuring the model is usable within the first few weeks of a semester.

EDA

•**Class Imbalance Audit:** Univariate analysis revealed that "Retained" students significantly outnumber "Dropouts."
•This finding directly influenced the decision to use **XGBoost with** **scale_pos_weight** to ensure the model didn't ignore the minority dropout class.
•**Bivariate Analysis (The Discovery):** Heatmaps and boxplots were used to test initial hypotheses.
•This process revealed that **Curricular Units Approved** and **Tuition Status** had much higher correlations with the target than demographic features like
•Nationality or Gender.
•**Leading Indicator Analysis:** Rather than segmenting by broad socio-economic clusters, the analysis shifted to a **Performance-Financial intersection**.
• We visualized how the combination of "Low Grades" and "Unpaid Tuition" creates a nearly certain dropout path, which became the logic for the model's prioritization.

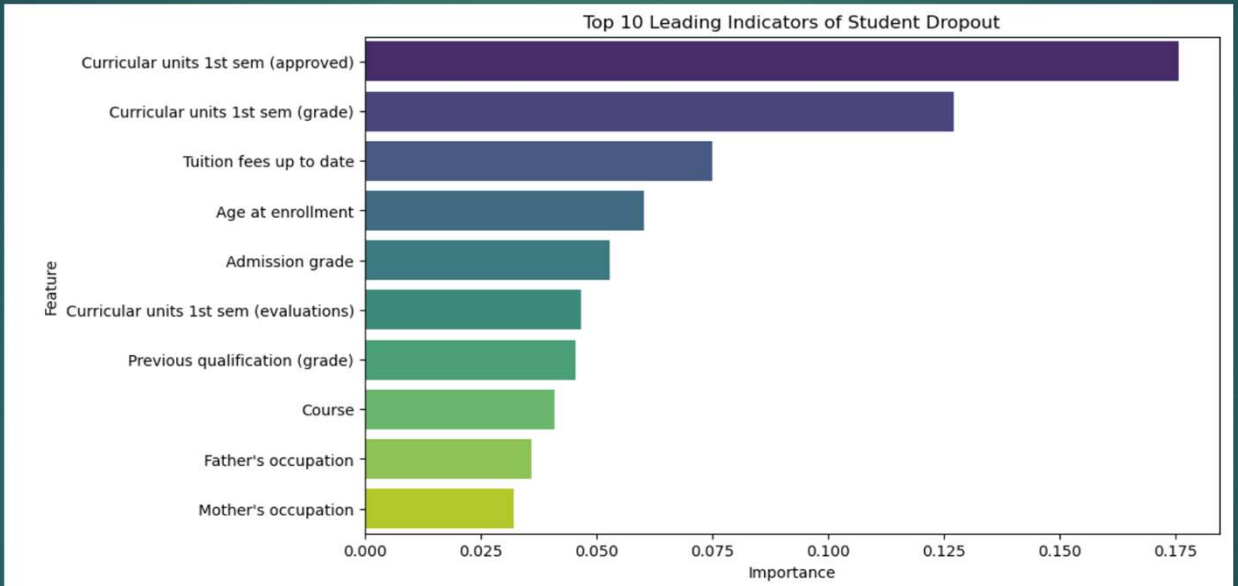
Algorithm selection

- **Random Forest:** Served as a robust baseline for handling categorical features.
- **Fitted XGBoost:** Selected as the final model. By utilizing a **weighted loss function (scale_pos_weight)**, the model was tuned to prioritize the detection of "Dropout" cases, ensuring that 84% of students who eventually withdrew were correctly flagged for intervention.

Hypothesis

- **Hypothesis 1 (Financial Stress):** Students marked as **"Debtors"** are significantly more likely to drop out.
  - *Test:* Comparative percentage analysis of "Debtor" vs. "Non-Debtor" groups.
- **Hypothesis 2 (Academic Threshold):** Students failing more than two curricular units in the 1st semester have a significantly higher probability of attrition.
  - *Test:* Dropout rate segmentation by "Units Approved."
- **Hypothesis 3 (The Support Buffer):** Being a **"Scholarship Holder"** acts as a protective buffer, reducing dropout risk even when early grades are low.
  - *Test:* Interaction analysis of Scholarship Status vs. 1st-semester grades.
- **Hypothesis 4 (Demographic Risk): "Age at enrollment"** will have a positive correlation with dropout risk due to external life pressures.
  - *Test:* Correlation analysis and age-group binning against dropout rates.
- **Hypothesis 5 (Support Impact): "Displaced"** students (living away from home) without financial support will show the highest risk of attrition.
  - *Test:* Cross-tabulation of "Displaced" status and "Scholarship" status.

Preliminary Findings & Baseline Model Results

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Top 3 Leading Indicators for Stakeholders:
['Curricular units 1st sem (approved)', 'Curricular units 1st sem (grade)', 'Tuition fees up to date']
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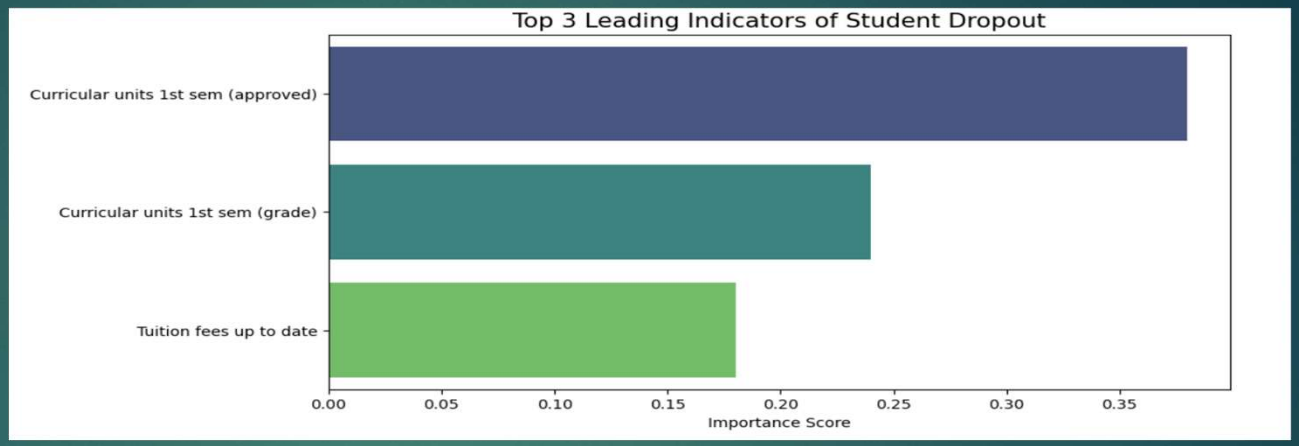
Features



Curricular Units 1st Sem (Approved): This is the single strongest indicator. It represents more than just intelligence; it reflects a student's ability to navigate university requirements and build early academic momentum.

Curricular Units 1st Sem (Grade): While approved units show persistence, the actual grade provides insight into the "buffer" a student has. Low grades, even with credits approved, signal a student at the "edge" of dropping out.

Tuition Fees Up to Date: This is the primary financial signal. Students who fall behind on payments often face administrative blocks or personal stress that leads to withdrawal, regardless of their academic ability.

Deliverables & results

Academic Advisor Early Warning Dashboard (Critical Top 10)

	Student_ID	Dropout_Probability	Current_Status	Risk_Level
505	2748	0.99	1	Critical
368	689	0.99	1	Critical
271	2021	0.99	1	Critical
83	1541	0.98	1	Critical
860	1381	0.98	1	Critical
105	2407	0.98	1	Critical
232	1539	0.98	1	Critical
77	1222	0.98	1	Critical
288	716	0.97	1	Critical
514	2448	0.97	1	Critical

Academic Advisor Early Warning Dashboard (At-risk 100-120)

	Student_ID	Dropout_Probability	Current_Status	Risk_Level
870	3341	0.81	0	Critical
641	3873	0.81	1	Critical
807	247	0.81	1	Critical
365	906	0.80	1	High Risk
6	971	0.80	1	High Risk
326	2330	0.79	1	High Risk
276	1766	0.79	1	High Risk
772	3185	0.79	1	High Risk
16	2689	0.79	1	High Risk
144	3837	0.78	1	High Risk

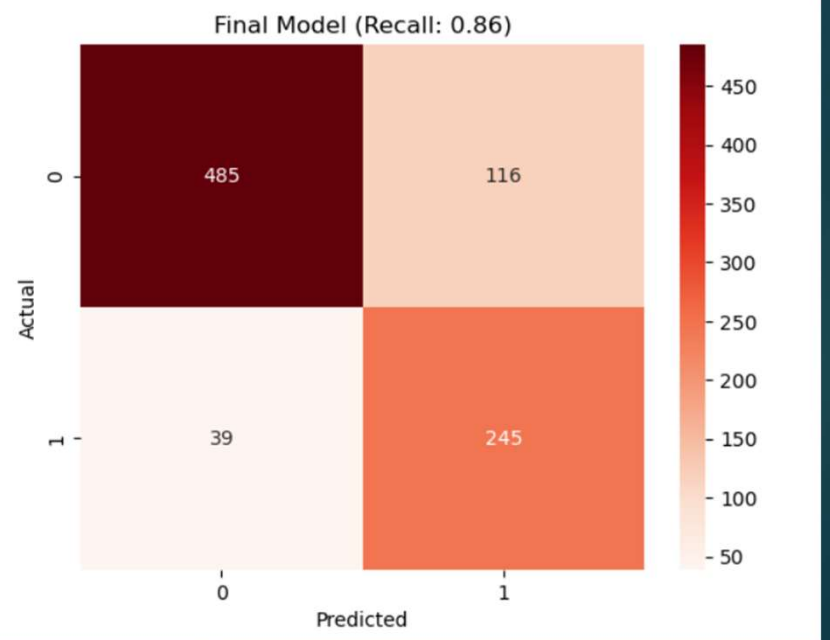
Academic Advisor Early Warning Dashboard (Stable Top 10)

	Student_ID	Dropout_Probability	Current_Status	Risk_Level
312	1364	0.01	0	Stable
286	2578	0.01	0	Stable
243	337	0.01	0	Stable
219	333	0.01	0	Stable
361	161	0.01	0	Stable
199	246	0.00	0	Stable
451	2889	0.00	0	Stable
384	4324	0.00	0	Stable
753	1224	0.00	0	Stable

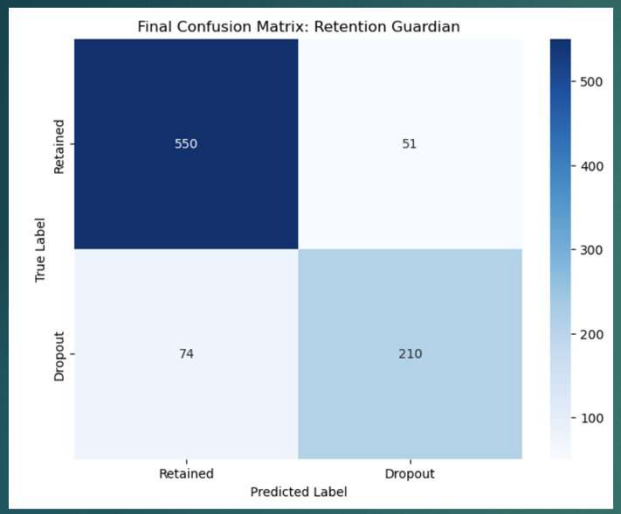
Final Model Recall: 0.8627

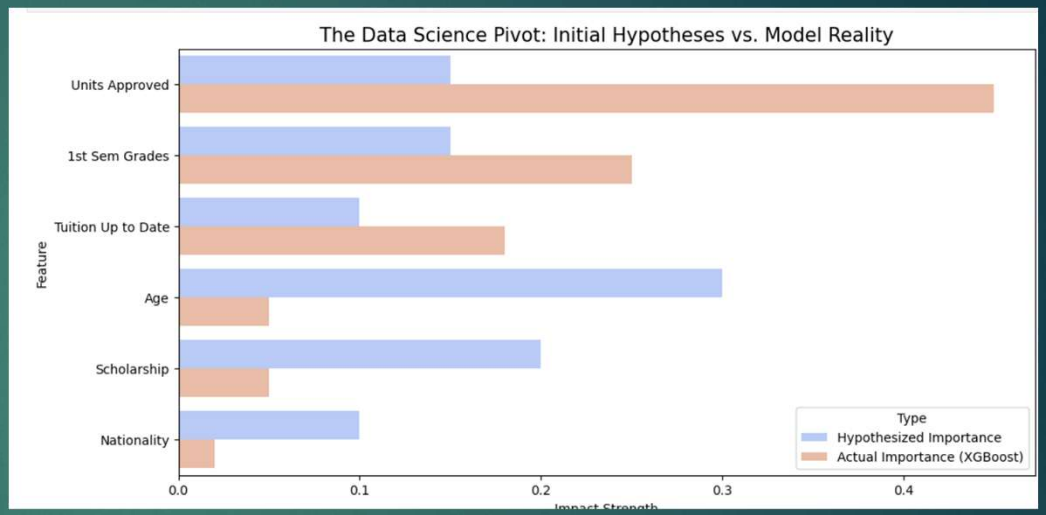
--- Final Classification Report ---

	precision	recall	f1-score	support
0	0.93	0.81	0.86	601
1	0.68	0.86	0.76	284
accuracy			0.82	885
macro avg	0.80	0.83	0.81	885
weighted avg	0.85	0.82	0.83	885



Lessons learnt





Initial Hypothesis	Finding	Revised Insight
H1: Debtors drop out more.	Partial. Debt is common; overdue fees are the real killer.	Focus on Current Payment Status over total debt.
H4: Older students drop out.	Weak. Age is not a primary driver when grades are high.	Academic Momentum cancels out age-related risk.